

Time Series and Forecasting Report

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1 Introduction

Time series analysis is a crucial tool in understanding and predicting temporal data. In this report we will do an exploratory data analysis, then we will model it and to do some forecasting in a time series that we choose to analyse. Due to the interest of stocks and trading around the world we will analyse and explore the historical stock price for S&P 500 Index using Yahoo Finance databases.

Stock prices move up and down all the time, and having the ability to predict its movement has immense potential to make one rich. Stock price prediction has kept people interested for a long time. There are hypotheses like the Efficient Market Hypothesis, which says that it is almost impossible to beat the market consistently, and others disagree with it.

2 Dataset

The historical stock price information is publicly available. In this case we will use `Yahoo_fin`, a python package [1], designed to scrape historical stock price data, as well as to provide current information on market caps, dividend yields, and which stocks comprise the major exchanges.

With this package, we are able to extract data from 1970 to the current day in the format presented in Fig. 1. Until the day of this report is being written we have 13519 observations.

	open	high	low	close	adjclose	volume
1970-03-25	0.0	91.070000	88.110001	89.769997	89.769997	17500000
1970-03-26	0.0	90.650002	89.180000	89.919998	89.919998	11350000
1970-03-30	0.0	90.410004	88.910004	89.629997	89.629997	9600000
1970-03-31	0.0	90.169998	88.849998	89.629997	89.629997	8370000
1970-04-01	0.0	90.620003	89.300003	90.070000	90.070000	9810000

Figure 1: Dataset format

As seen in Fig. 1, we have seven attributes: "date", "open", "high", "low", "close", "adjclose" and "volume". The first attribute is the trading Date, i.e. the day that information is related to. The second and fifth are the prices at which a stock began and ended trading in the same period. The third and fourth refer to the maximum and minimum prices in a given time period. The sixth is the adjusted values factor in corporate actions such as dividends, stock splits, and new share issuance. The last is the total amount of trading activity.

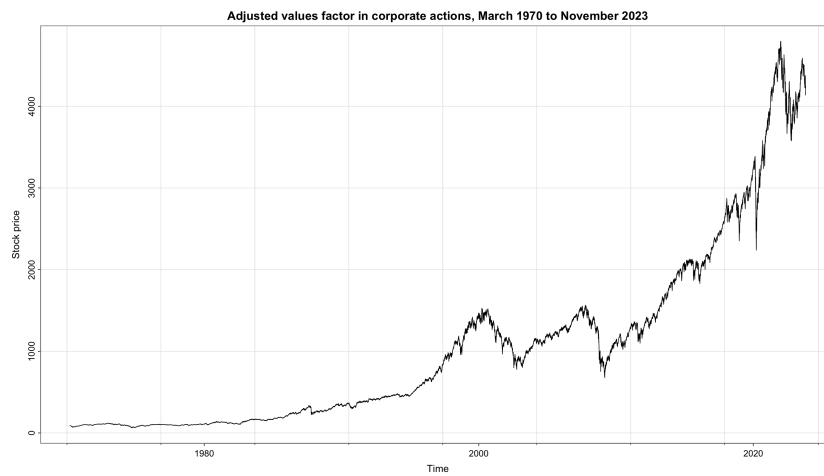
3 Data visualization

For the purpose of our work we will only consider the "adjclose" variable to analyse since it's the most used variable in terms of analysing this type of data. We choose to analyze this variable because by using the adjusted value of the data we also take in consideration the dividends, stock splits, and new share issuance making our analysis more robust overall.

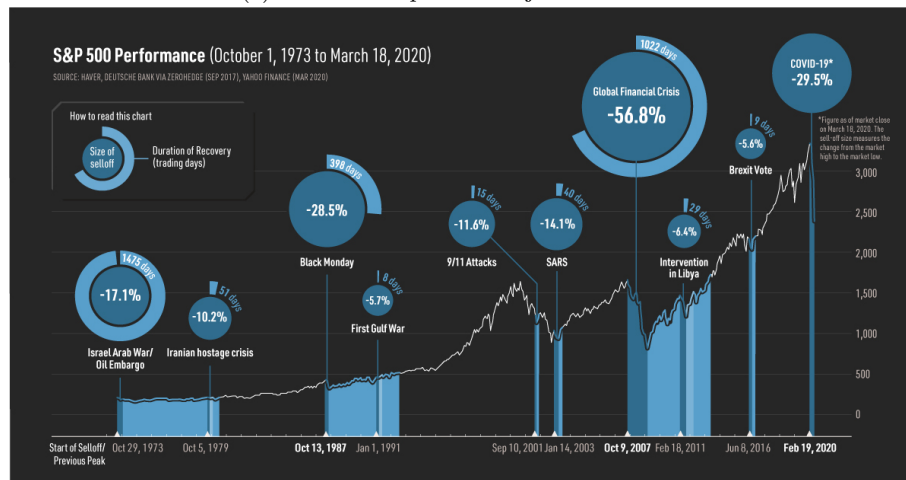
With that, the initial steps that we need to do is to visualize the graph the data of "adjclose" along time. In the Fig. 2a we can see the time series plot of that variable. We can say that there has been a gradual increase in the stock price since 1970 to the present day with some drops in between, but not very significant in the general analysis. However, there are three sharp downfalls in 2001, 2007 and in 2020. These sharp downfalls can be explained by analyzing the Fig. 2b. This leads us to say that this time series has a trend, and is not always increasing or decreasing, so it looks a little bit random.

4 Correlation

Since the time series has a trend, it's not stationary, and the ACF (Autocorrelation Function) does not make sense, but we can always compute it. In Fig. 3a, we can see the ACF plot, so we can conclude that it reflects the trend since it is decaying very slowly, as we can see in Fig. 3b. This means that the correlation between successive observations is very high, nearly 1, as seen in Fig. 3b. These plots seem to indicate that the time series does not present seasonality. Furthermore, as can be seen in Fig. 3a, the significance threshold is growing over the number of lags. The reason for that is we are using Python to explore this time series and to create all the plots, and this software, contrary to, for example, in Bartlett test, the formula $\pm \frac{1.96}{\sqrt{n}}$ to calculate the significance threshold and it remains all the same, python uses a different approach on calculating this. It doesn't directly get significance thresholds. Instead, it typically gets the autocorrelation values at various lags. To determine if those autocorrelation values are statistically significant, and would typically perform hypothesis tests or compare them to critical values based on a specific statistical distribution.

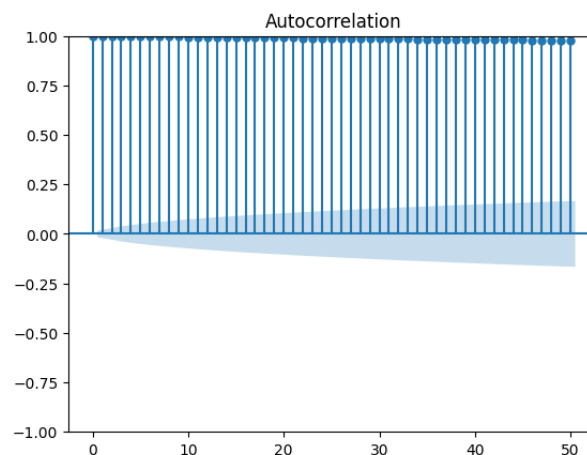


(a) Time series plot of "adjclose" variable



(b) S&P 500 Performance

Figure 2: Time-series plots



(a) ACF plot

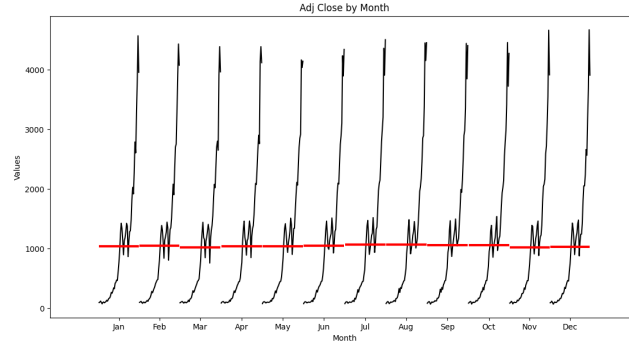
```
[1.      0.99955689 0.99913188 0.99868985 0.99823981 0.99780383
 0.99736172 0.99692608 0.99646338 0.99600889 0.9955303  0.99506896
 0.99459972 0.99412194 0.99365558 0.99318682 0.99273914 0.99229025
 0.99184597 0.99140283 0.99094867]
```

(b) ACF values with 20 lags

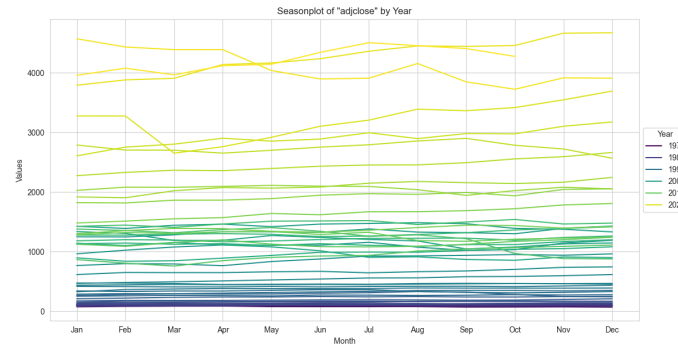
Figure 3: ACF of the "adjclose" variable

5 Seasonality and month plots

From previous analyses is not obvious if the time series has seasonality or not. What we can do is to have a look into Fig. 4a and Fig. 4b. When analyzing them, these plots do not tell us much about seasonality. However, these indicate that the time series seems to not present seasonality.



(a) Month plot



(b) Season plot

Figure 4: Seasonality plots for our time series

6 Analysing the returns

Before going into any further analysis, we need to know what is the returns. The return measures the relative gain or loss from day to day. Our particular interest here is the return or relative gain or loss in a trading day. Analyzing stocks through the lens of returns plays a fundamental role in understanding the dynamic nature of financial markets. This approach quantifies the daily fluctuations in stock prices, offers a unique perspective on stock behavior, and is a crucial step in financial data analysis. Returns provide insights into the inherent volatility, trends, and patterns exhibited by stocks, enabling the construction of predictive models.

	open	high	low	close	adjclose	volume
count	1.047400e+04	10474.000000	10474.000000	10474.000000	10474.000000	1.047400e+04
mean	inf	0.000382	0.000398	0.000409	0.000409	inf
std	NaN	0.008946	0.010473	0.011403	0.011403	NaN
min	-1.000000e+00	-0.131164	-0.201371	-0.204669	-0.204669	-1.000000e+00
25%	-4.386073e-03	-0.003496	-0.004156	-0.004516	-0.004516	-8.699827e-02
50%	6.773932e-04	0.000433	0.000789	0.000564	0.000564	2.768125e-04
75%	5.607037e-03	0.004581	0.005335	0.005718	0.005718	9.990161e-02
max	inf	0.080379	0.094105	0.115800	0.115800	inf

Figure 5: Statistical measures of returns.

If x_t is the value of the S&P 500 index on trading day t , then $r_t = (x_t - x_{t-1})/x_{t-1}$ is called the return. Typically r_t presents periods of high volatility clustered together. In the Fig 5, we can the statistical measures of the returns.

In the Fig 6 we present the daily trading returns, and we can see at least three significant pics. These are related to crisis in the world, and this off course influentiates the stock price. Closely, in 1987 occurred the "Black Monday"; in 2007-2008 occurred the "Global Financial Crisis"; in 2020 appeared COVID-19.

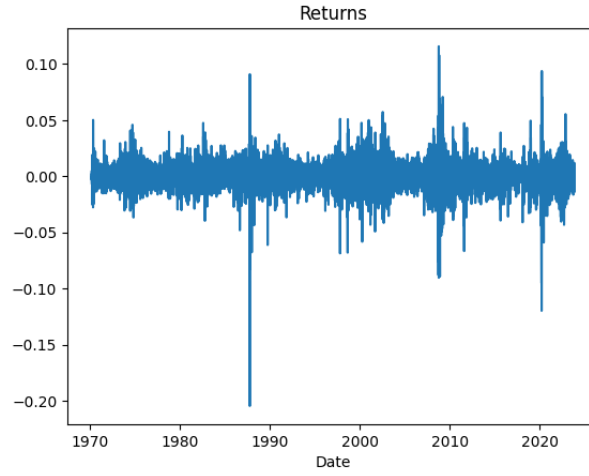


Figure 6: Plot of the returns.

6.1 Correlation

Now, we can have a look into the ACF and PACF of the returns if there is any evidence of correlation.

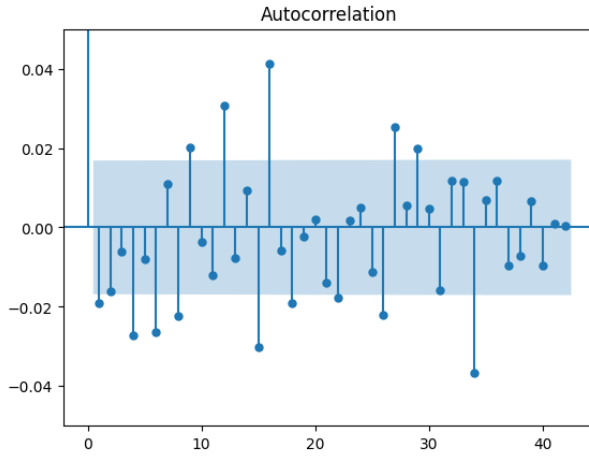


Figure 7: ACF of the returns.

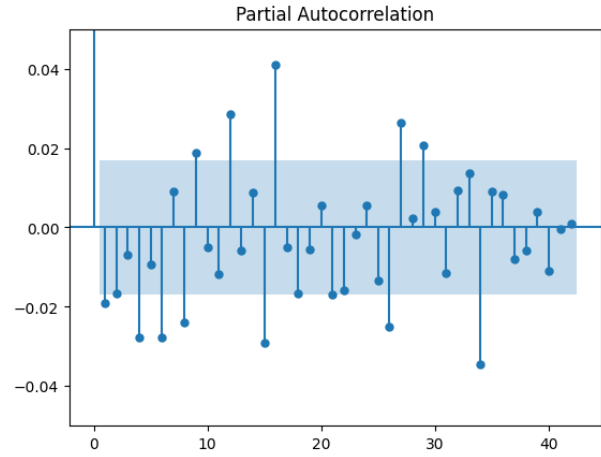


Figure 8: PACF of the returns.

As we can see in the Figs 7, 8 r_t presents minimal autocorrelation, but one particular property is that the r_t^2 has strong serial correlation.

In Fig 9 we can see the plot of the squared returns over time. Similarly to returns, the squared returns have three significant pics, as we indicated before. Now, we can have a look into the ACF and PACF of the squared returns.

r_t have minimal autocorrelation, but by seeing the Figs 10, 11 r_t^2 have serial correlation - this indicates that there are moments, notice that in serial correlation is related to a set of order moments, but if we do the squared of the returns we have fourth order moments in the returns, and the existence of higher order moments is linked to non-linear models. So, this is an indication that we need to go further than the linear models and the normal distribution.

By seeing the Fig 12 we conclude that is a familiar shape! It turns out that returns follow an almost normal distribution, centered on 0.

What we want to model is the variance of the return. The variance of return depends on the past variance of returns. Volatility is the measure of the spread of these returns around the mean. Volatility provides a measure of how much the values of a financial instrument, such as a stock or an index, fluctuate over a specific period. Higher volatility implies larger price fluctuations, indicating greater risk or uncertainty. On the other hand, lower volatility suggests more stable and predictable price movements.

7 Modeling the Data

In this section, we delve into the process of model selection. Our goal is to identify the model that best captures the dynamic patterns within the S&P 500 index, offering reliable insights into the factors influencing stock returns. Through a comprehensive evaluation, we aim to pinpoint the optimal model that aligns with market intricacies and serves as a robust tool for forecasting in the ever-changing landscape of financial markets.

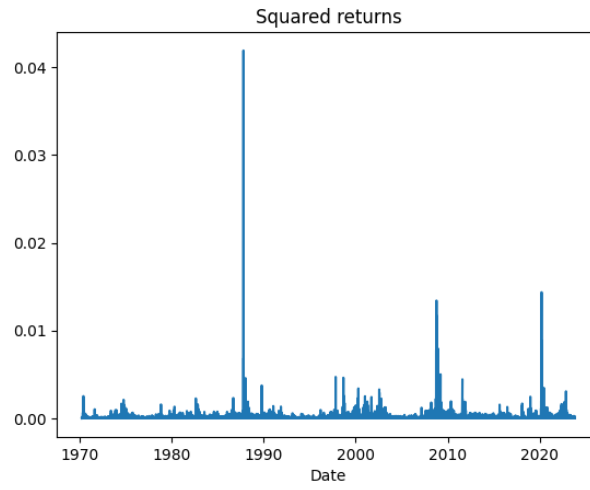


Figure 9: Plot of the squared returns.

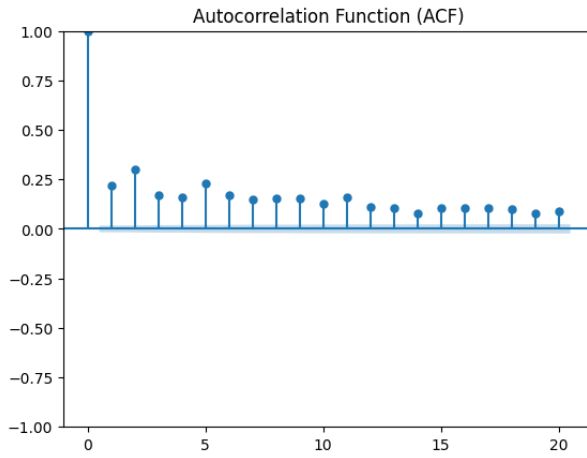


Figure 10: ACF of the squared returns.

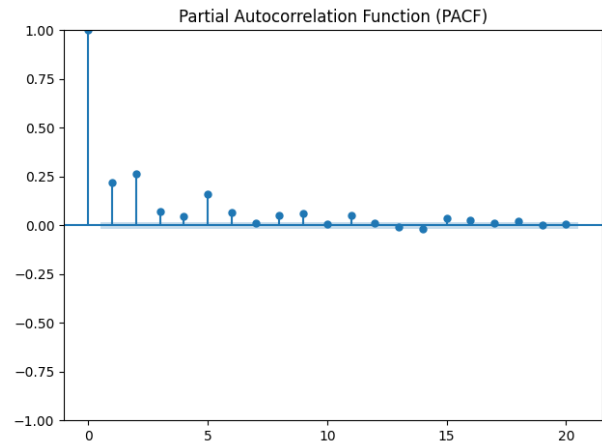


Figure 11: PACF of the squared returns.

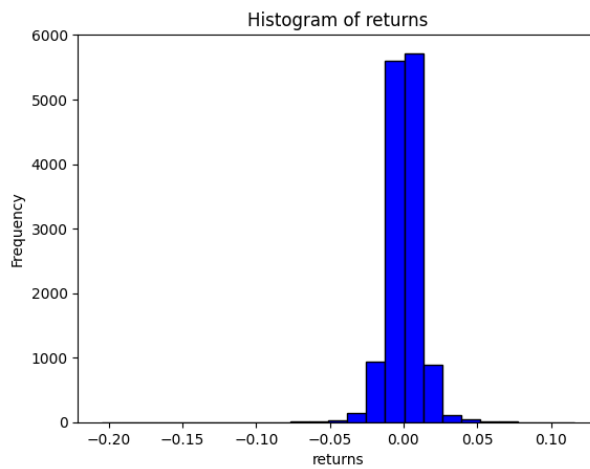


Figure 12: Histogram of returns.

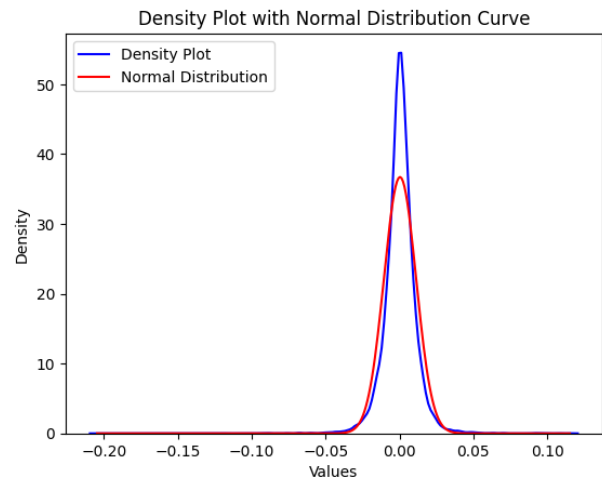


Figure 13: Density Plot of the returns with Normal Distribution Curve.

ARCH and GARCH models were developed in the eighties to model changes in volatility. The GARCH model is a generalization of the ARCH model that allows more complex dependence structures. Therefore, we need to choose one of these two models. The P/ACF of the squared returns from an ARCH(p) model will look like an AR(p). However, if the P/ACF appears to have an ARMA structure, the GARCH model is used. The Figs 10, 11 indicate ARMA autocorrelation, so a GARCH will be used.

7.1 Training the Model

The model training process involved a judicious split of the dataset into a training set, encompassing 80% of the historical data, and a test set, comprising the remaining 20%. This division is crucial for evaluating the model's generalization performance on unseen data. The training set serves as the foundation for the model to learn the underlying patterns, while the test set acts as an independent benchmark to assess how well the model extrapolates its learned insights to new observations. The use of an 80-20 train-test split strikes a balance between providing the model with sufficient information to capture intricate patterns and ensuring robust evaluation on unseen data, thereby mitigating the risk of overfitting. This methodology aligns with best practices in machine learning and time series analysis, enabling a comprehensive assessment of the model's predictive capabilities and its potential utility in forecasting future S&P 500 index returns.

Since the returns presents minimal autocorrelation, seen in Figs 7, 8, we need to fit an AR model or a MA model. Recall that in this case we will consider AR model since MA works as well and the same. We choose AR(2), and it can be seen the summary of the model in Fig 14.

SARIMAX Results						
=====						
Dep. Variable:	adjclose	No. Observations:	10829			
Model:	ARIMA(2, 0, 0)	Log Likelihood	-16235.399			
Date:	Tue, 16 Jan 2024	AIC	32478.797			
Time:	21:35:19	BIC	32507.957			
Sample:	0	HQIC	32488.629			
	- 10829					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]
=====						
const	0.0320	0.011	3.031	0.002	0.011	0.053
ar.L1	0.0118	0.005	2.507	0.012	0.003	0.021
ar.L2	-0.0396	0.004	-9.917	0.000	-0.047	-0.032
sigma2	1.1742	0.005	231.552	0.000	1.164	1.184
=====						
Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	168442.82			
Prob(Q):	0.99	Prob(JB):	0.00			
Heteroskedasticity (H):	2.22	Skew:	-0.67			
Prob(H) (two-sided):	0.00	Kurtosis:	22.27			
=====						

Figure 14: AR(2) model summary.

The AR(2) model results provide valuable insights. The coefficients reveal that the lag-1 (ar.L1) and lag-2 (ar.L2) autocorrelations are 0.0118 and -0.0396, respectively. The statistically significant p-values for these coefficients ($p=0.012$ for ar.L1 and $p<0.01$ for ar.L2) lend support to the model's efficacy in capturing the autocorrelation patterns in the data.

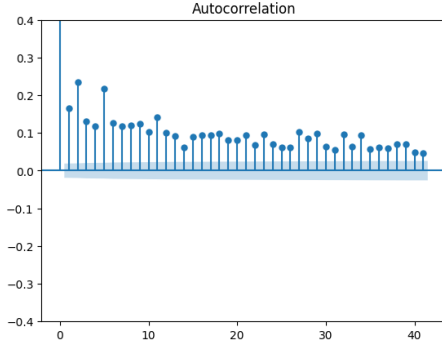


Figure 15: ACF of the squared residuals of AR(2) model.

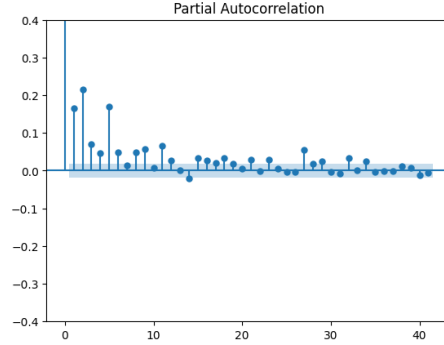


Figure 16: PACF of the squared residuals of AR(2) model.

Upon examining the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots of the squared residuals from the AR(2) model, presented in Figs 15, 16 we can conclude that they indicate ARMA autocorrelation.

Incorporating a GARCH model with the AR(2) model enhances our analysis by addressing the inherent limitations of traditional autoregressive models in capturing time-varying volatility, we used the function `arch_model` from package `arch` [2]. While the AR(2) model adeptly captures autocorrelation patterns in squared returns, financial time series often exhibit fluctuations in volatility over time. The GARCH model, designed to capture conditional heteroskedasticity, introduces lagged conditional variances, providing a more nuanced representation of volatility dynamics. This approach is particularly beneficial in financial markets where periods of high and low volatility cluster. By combining the AR(2) model for autocorrelation with the GARCH model for capturing time-varying volatility, our framework becomes more adept at forecasting and comprehensively understanding the complex dynamics within the S&P 500 index returns, ultimately leading to improved accuracy and reliability in our predictions.

The AR(2)-GARCH(1,1) model results provide valuable insights into the dynamics of volatility within the S&P 500 index returns. The mean model indicates a statistically significant constant mean (μ) of 0.9197, highlighting a baseline level of volatility in the absence of specific market events. Moving to the volatility model, the GARCH(1,1) structure is employed, featuring an omega term

(0.0104) capturing the long-term volatility level, an alpha term (0.0715) representing the short-term impact of past squared returns, and a beta term (0.8985) depicting the persistence of past volatility shocks. The model's R-squared and adjusted R-squared, while both at zero, are expected in GARCH models where the focus is on capturing volatility rather than predicting the mean. The log-likelihood and information criteria (AIC and BIC) indicate a reasonable fit of the model to the data. Overall, the GARCH model provides a comprehensive framework for modeling volatility, offering valuable insights into the conditional heteroskedasticity present in the S&P 500 index returns, and enhances our ability to forecast future volatility patterns in financial markets.

AR - GARCH Model Results					
Dep. Variable:	adjclose	R-squared:	-0.003		
Mean Model:	AR	Adj. R-squared:	-0.003		
Vol Model:	GARCH	Log-Likelihood:	-14264.8		
Distribution:	Normal	AIC:	28541.7		
Method:	Maximum Likelihood	BIC:	28585.4		
Date:	Tue, Jan 16 2024	No. Observations:	10827		
Time:	22:44:47	Df Residuals:	10824		
	Mean Model	Df Model:	3		
	coef	std err	t	P> t	95.0% Conf. Int.
Const	0.0477	8.072e-03	5.910	3.420e-09	[3.188e-02,6.352e-02]
adjclose[1]	0.0700	1.010e-02	6.931	4.183e-12	[5.018e-02,8.975e-02]
adjclose[2]	-0.0135	1.059e-02	-1.272	0.203	[-3.422e-02,7.287e-03]
Volatility Model					
	coef	std err	t	P> t	95.0% Conf. Int.
omega	0.0104	2.688e-03	3.867	1.101e-04	[5.126e-03,1.566e-02]
alpha[1]	0.0715	1.231e-02	5.812	6.167e-09	[4.742e-02,9.568e-02]
beta[1]	0.9197	1.248e-02	73.684	0.000	[0.895, 0.944]

Figure 17: AR(2)-GARCH(1,1) model summary.

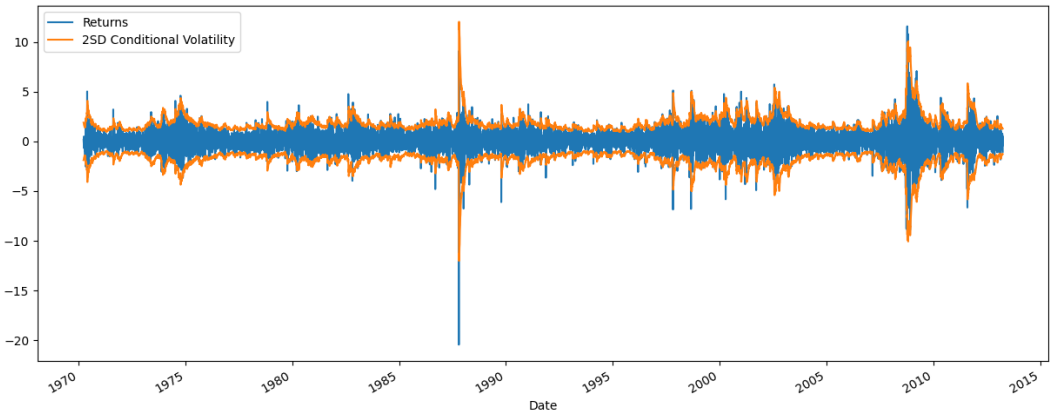


Figure 18: Fitted values

By seeing the Fig 18, our model demonstrated a commendable fit to the S&P 500 index returns data, effectively capturing both autocorrelation patterns and time-varying volatility.

7.2 Model residuals

The residual analysis of our model serves as a critical step in validating the model's performance and assessing its ability to account for unforeseen patterns within the S&P 500 index returns. By examining the residuals, the discrepancies between the model's predictions and the actual data can be scrutinized, shedding light on any remaining systematic patterns or information that may not have been fully captured. This analysis is pivotal in ensuring the model's robustness, providing insights into its limitations, and guiding potential refinements. The objective of the residual analysis is to evaluate whether the model adequately accounts for autocorrelation, heteroskedasticity, and any other structures in the data, ultimately validating the model's reliability for forecasting and decision-making.

The residuals exhibit a favorable pattern, presented in Fig 19, characterized by a consistent mean and stable spread over time, signifying that our model effectively captures the underlying dynamics of the data.

The ACF and PACF of the residuals, presented in Figs 20, 21, display minimal correlation, indicating that our model effectively captures and accounts for the autocorrelation patterns within the data, thus validating its suitability for forecasting.

Upon inspecting the normality of the residuals from the model, it became evident that they deviate from a normal distribution as we can see in the Fig 22. This departure from normality suggests that the model might not entirely capture all the underlying

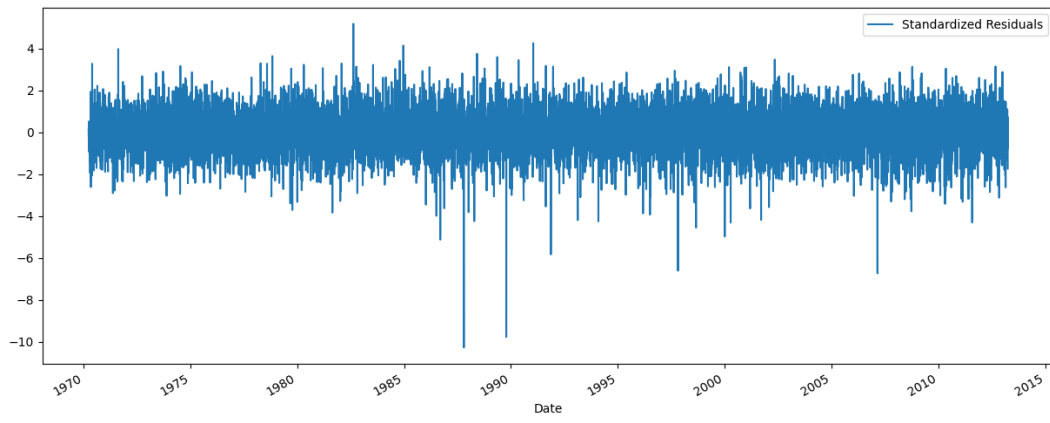


Figure 19: Model Residuals

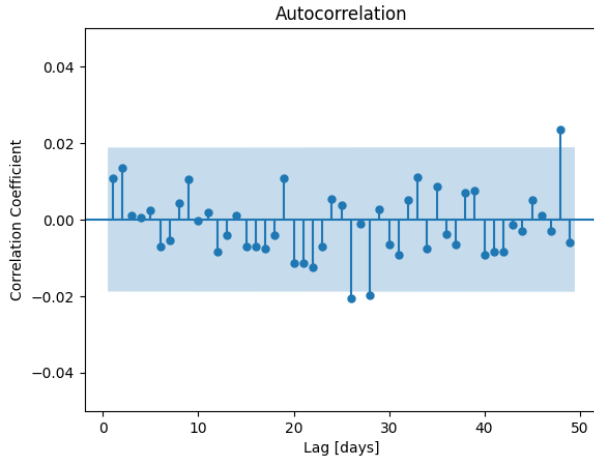


Figure 20: ACF of the model residuals.

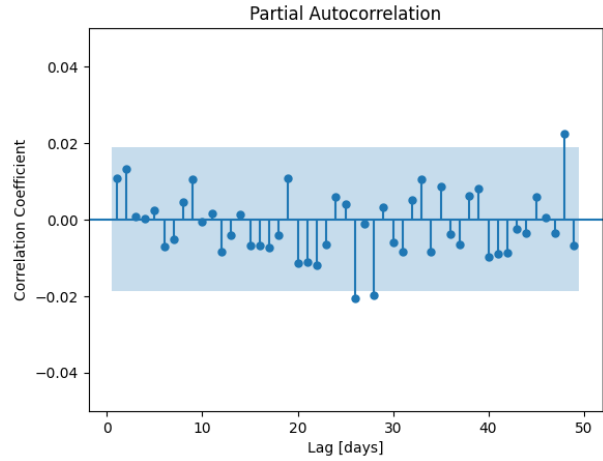


Figure 21: PACF of the model residuals.

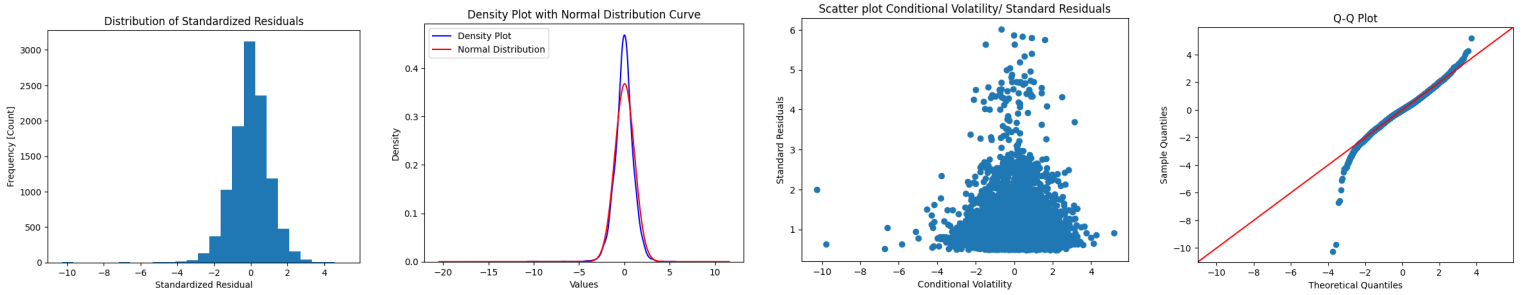


Figure 22: Residual plots of Model

complexities within the data, as residual normality is an essential assumption for statistical inference. Non-normal residuals could imply the presence of unaccounted factors, outliers, or potential misspecifications in the model structure.

7.3 Analyzing the results

The predictions derived from the model exhibit a remarkable level of accuracy when compared to the test data, underscoring the model's robustness in forecasting. The consistency between the model's forecasts and the actual test data reflects a high degree of reliability, signifying that the model effectively captures the underlying dynamics of the financial time series.

7.4 Forecasts and Confidence intervals

In the evaluation of confidence intervals for our AR(2)-GARCH(1,1) model, we employed both Gaussian and bootstrapping methods to assess the reliability and robustness of our parameter estimates. Remarkably, at a 95% confidence level, the Gaussian method consistently yielded smaller confidence intervals compared to the bootstrapping technique as showed in Fif26. The Gaussian method assumes normality in the parameter estimates, aligning with traditional statistical assumptions. This led to more tightly

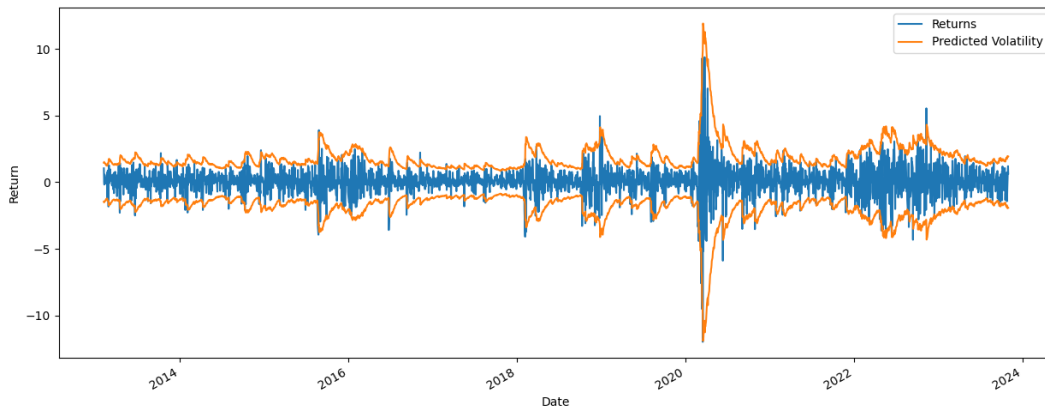


Figure 23: Model predictions

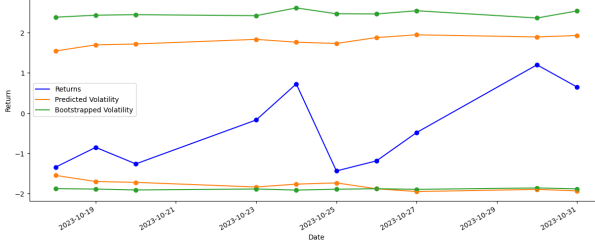


Figure 24: Gaussian and Bootstrapped Confidence Intervals

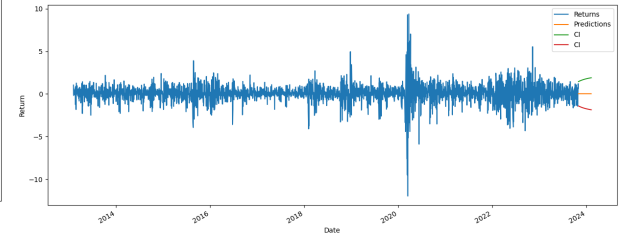


Figure 25: Model Predictions with Confidence Intervals of Model

constrained intervals, indicative of higher precision in the parameter estimates. In contrast, the bootstrapping method, being a non-parametric resampling approach, may be less restrictive in assuming specific distributional properties, resulting in wider confidence intervals. This discrepancy in interval widths underscores the sensitivity of our parameter estimates to the underlying distribution assumptions and the potential impact on the precision of our model. While the Gaussian method provides narrower intervals, caution is warranted, and future work could explore the implications of these differences in the context of the dynamic and non-linear nature of financial time series data.

observed	high_boot	low_boot	high_gaus	low_gaus
-0.009824505308242415	2.38294177845985	-1.8736651256970371	1.608705084179502	-1.608705084179502
-1.3399820506516447	2.43289620154226	-1.884231796370189	1.5452398763717188	-1.5452398763717188
-0.8482848196321058	2.4455448000187605	-1.9064017451679593	1.696132193338447	-1.696132193338447
-1.2585283719027562	2.4213975331198685	-1.8822623989899667	1.7181032877604316	-1.7181032877604316
-0.16855698941634634	2.6137734334566947	-1.9094733919112186	1.8328525109672098	-1.8328525109672098
0.7265792222727274	2.468617725543937	-1.8882869412293697	1.761856647437843	-1.761856647437843
-1.4339628627712542	2.463478160861659	-1.8756648486621912	1.7312187554078964	-1.7312187554078964
-1.1832519778109618	2.543295053776157	-1.8912488257042397	1.8775420868908672	-1.8775420868908672
-0.48002802297685276	2.363196222973069	-1.8564723369742908	1.9453360192506106	-1.9453360192506106
1.2010022325859904	2.53902719783707	-1.878881605190772	1.892411368269409	-1.892411368269409

Figure 26: Gaussian and bootstrapped confidence intervals

8 Transforming the Data

Addressing the non-normality observed in the residuals of the model prompts a critical exploration of potential data transformations. These transformations seek to mold the distribution of residuals towards normality, offering a pathway to enhance the model's accuracy and reliability.

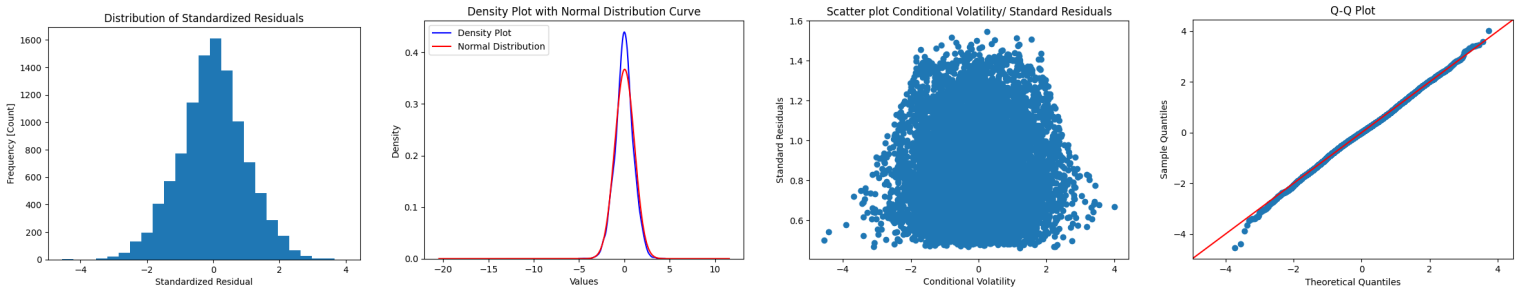


Figure 27: Residual plots of Model after removing outliers

8.1 Removing Outliers

The removal of outliers, while visibly improving the normality of the residuals Fig 27, resulted in a diminished performance of the model. This outcome underscores the delicate balance in handling outliers, as their exclusion may lead to a loss of critical information essential for capturing the inherent complexities within the data as seen in Figs 28, 29 and 30.

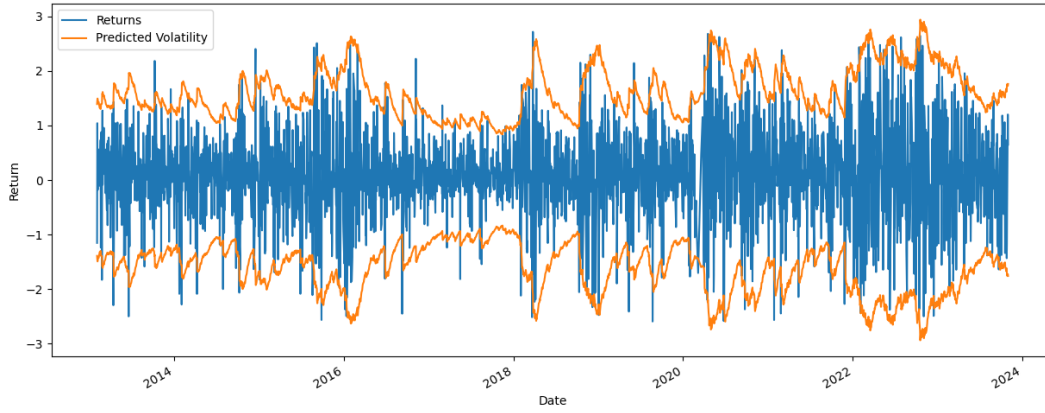


Figure 28: Model predictions of Model after removing outliers

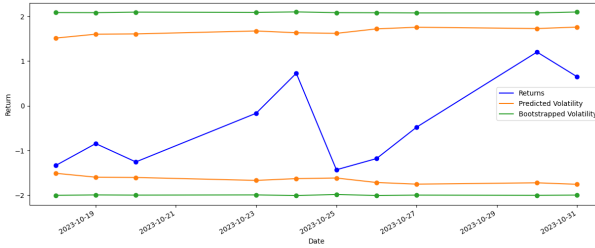


Figure 29: Gaussian and Bootstrapped Confidence Intervals

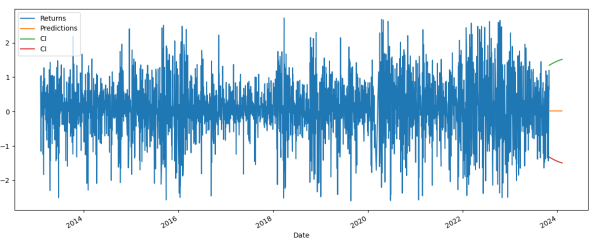


Figure 30: Model Predictions with Confidence Intervals

8.2 Logarithm of Data

The application of the log transformation to the residuals of the model did not yield the anticipated improvement in normality. Despite the log transformation's ability to reduce skewness and potentially stabilize variance, the resulting distribution of residuals remained noticeably non-normal as seen in Fig31. This outcome suggests that the log transformation might not be the most suitable approach for addressing the non-normality observed in the financial time series data.

8.3 Box-Cox

The application of the Box-Cox transformation to the residuals of the model did not result in a substantial improvement in achieving normality. Despite the flexibility of the Box-Cox method, which encompasses a range of power transformations, the transformed residuals exhibited persistent non-normal characteristics as seen in Fig32. This observation suggests that, in this data, the Box-Cox transformation may not be the most effective remedy for addressing non-normality in the residuals.

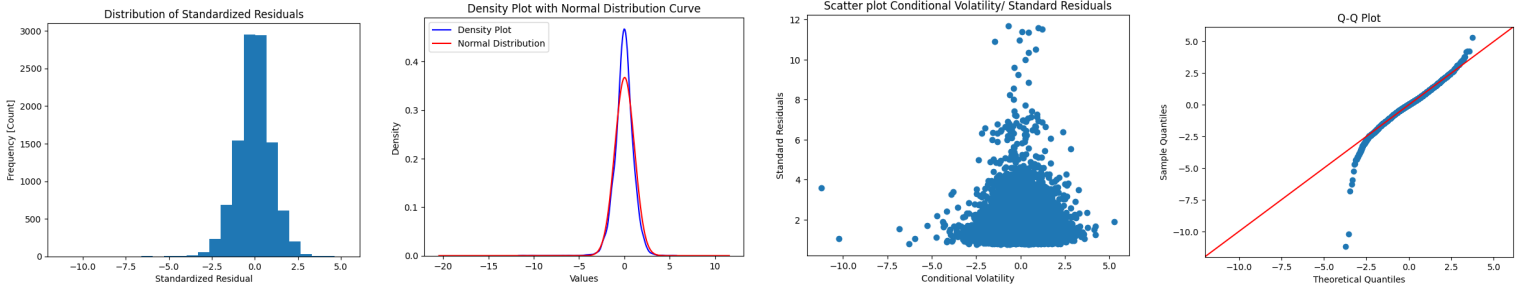


Figure 31: Residual plots of Model of log of the returns.

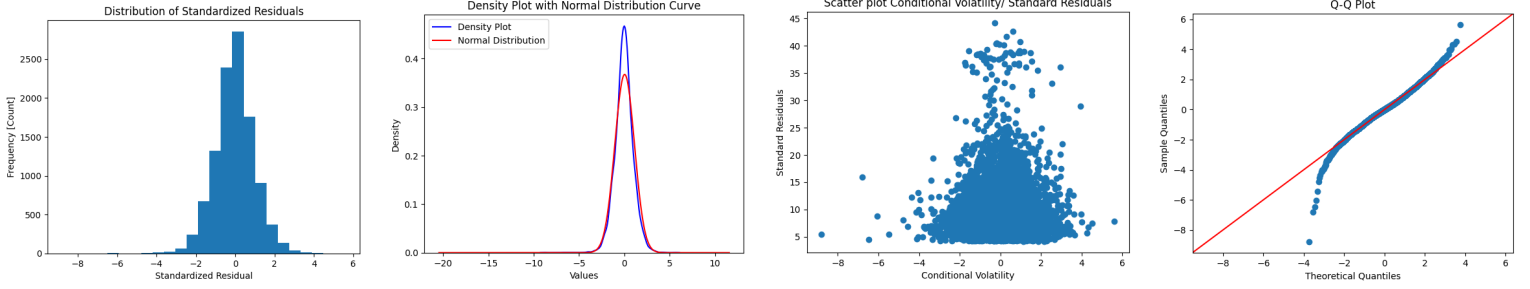


Figure 32: Residual plots of Model of Box-Cox of the returns

9 Comparison

We built four models to find the most suitable one. First, using AR(2)-GARCH(1,1) with the value of the returns we got a nice model that predicted and forecast pretty well. In the second model, we saw that in the first model, the normality of the residuals was little bit comprised, and the reason why is because of the presence of outliers. So, we decided to remove some outliers to see if there was the case, and it was, there was some cases that influentiated negatively the normality of the residuals but with this data we can remove it because these outliers corresponded to import events in the world, like global crisis, wars, etc. In the third model and fourth model, we performed the logarithm of the returns and the box-cox transformation to try to normalize the residuals of the model, but it was a failure, since the residuals still were not normally distributed by seeing the Q-Q plot.

10 Conclusion

In conclusion, the AR(2)-GARCH(1,1) hybrid model emerged as the most robust framework for modeling and forecasting S&P 500 index returns, outperforming the other models tested in this analysis. Despite encountering challenges related to the non-normality of residuals, the AR(2)-GARCH(1,1) consistently demonstrated superior predictive accuracy compared to models subjected to transformative strategies. Notably, the attempted removal of outliers, log transformation, and Box-Cox transformation failed to effectively address the non-normality issue. While acknowledging the limitations in achieving normality, the empirical evidence supports the continued use of the AR(2)-GARCH(1,1) model due to its superior overall performance. Moving forward, the exploration of diverse models, such as machine learning approaches like neural networks or ensemble methods, may offer additional avenues for refining predictions and addressing the residual distribution concerns encountered in this analysis. The dynamic nature of financial markets requires an adaptable and comprehensive approach, prompting ongoing efforts to enhance our modeling strategies for more accurate and reliable forecasting.

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